DOB NOW Standard Energy Code Objections for 2020 NYCECC

Objection	Code Sections	Objection Details
EN-a [General]		
EN-a.1 DOB NOW/ EN Admin	DOB NOW	Indicate the **job application details** in DOB NOW to match the job documented on drawings.
EN-a.2 TR8	1 RCNY §5000-01 (h), 104.1, 104.2.1, 4.2.4	In **DOB NOW TR8** and **on drawings**, indicate the **required energy code progress inspections** to match the work scope of the job completely.
EN-a.3 Professional Statement	1 RCNY §5000-01 (e), 101.5.2.1	Provide a **professional statement** on EN- labeled sheet (a) Stating "... in compliance with the 2020 NYCECC" for ECC-compliance-path jobs, or (b) Stating "... in compliance with the 2020 NYCECC by application of ASHRAE 90.1-2016 as modified by Appendix CA" for ASHRAE-compliance-path jobs.
EN-a.4 Code Compliance Path	1 RCNY §5000-01 (d)	Follow a **single code compliance path** across all related job filings (initial, subsequent, and amendment filings), and throughout plans and energy analysis. (a) Residential building via ECC compliance path (Chapter R2 through R6), or (b) Commercial building via ECC compliance path (Chapter C2 through C6), or (c) Commercial building via ASHRAE compliance path (Appendix CA)
EN-a.5 Energy Analysis	1 RCNY §5000-01 (f), 101.5.2.2, 4.2.1.1; R501.4, C501.4, C501.7, C502.1, C503.1, C505.1, 4.2.1.2, 4.2.1.3	Provide complete **energy analysis** on EN- labeled sheet (a) Covering all proposed scope of work, and (b) All values and attributes reported in analysis clearly matching drawings. (c) Alterations-specific exceptions/allowances must be individually reported in Tabular Analysis with applicable code sections and supporting documentation locations.
EN-a.6 Envelope Diagrams	1 RCNY §5000-01 (g)	Provide **envelope diagrams** in elevation/plan format on EN- labeled sheet to support all envelope components reported in energy analysis. (a) Call out wall/fenestration/roof/floor types and surface area value (sf) of each type, and key them to the envelope energy analysis. (b) For Additions and Alterations to existing building envelope, indicate areas of the existing-to-remain, alteration, and new construction (addition) clearly in the envelope diagrams.
EN-a.7 Thermal Boundary Diagrams	1 RCNY §5000-01 (g)(5)(iv)	Provide **thermal boundary diagrams** on EN- labeled sheet to clarify thermal boundary of the proposed building envelope.
EN-a.8 Energy Modeling	1 RCNY §5000-01 (f)(iv), 1 RCNY §5000-01 (d)(3)(iii), 5.2.3; 11.5, G3.1	For jobs that chose **energy modeling** as energy analysis method: (a) Provide **EN1** on EN- labeled sheet -- all values and attributes reported in EN1 must clearly match drawings, and (b) Submit **model output reports** as a separate supporting documentation to validate entries in EN1, and (c) Submit a **supporting documentation index** to tabulate compliance with mandatory provisions. (d) For buildings ≥ 25,000 sf, provide **COMcheck envelope compliance certificate** to demonstrate compliance with additional envelope requirements.
EN-a.9 List of Progress Inspections	1 RCNY §5000-01 (h), 1 RCNY §5000-01 (g)(7)	Provide list of **required energy code progress inspections** on EN- labeled sheet. (a) Table I format for Residential building (b) Table II format for Commercial building

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EN-a.10 Supporting Documentation Index	1 RCNY §5000-01 (g)	Provide **supporting documentation index** on EN- labeled sheet in tabular format to list applicable code sections, to describe proposed energy design elements to meet the code, and to indicate document locations (drawing number and/or section title).
EN-a.11 Additional Efficiency Package Option	C406, Appendix I	Indicate the chosen **additional efficiency package option**, and provide specifications/details to support the efficiency option for a new Commercial building (or, new Tenant space).

Objection	Code Sections	Objection Details
EN-v [Envelope]		
EN-v.1 Exterior Wall Deductions (Zoning)	ZR 12-10 Definitions/Floor Area/(12)	To prove **exterior wall deductions eligibility**, provide calculation table to document: (a) Area-weighted average U-factor of all opaque above-grade wall assemblies in the proposed building is ≤ 80% of that in the code-prescribed building, and (b) Area-weighted average U-factor of all above-grade exterior wall assemblies, including vertical fenestrations, in the proposed building is ≤ 90% of that in the code-prescribed building. (c) For existing walls, all added wall thicknesses have minimum R-1.5/inch.
EN-v.2 (e) Maximum/ Minimum Fenestration Area	C402.4.1, C402.4.2; C503.3	For **ECC-compliance-path** jobs, provide **fenestration area calculation path** to prove (a) Vertical fenestration area ≤ 30%, Skylight area ≤ 3%, or (b) Vertical fenestration area ≤ 40% with Daylight responsive controls, Skylight area ≤ 6% with Daylight responsive controls, and (c) For spaces defined in C402.4.2, the Minimum skylight area provided
EN-v.3 (a) Maximum/ Minimum Fenestration Area	5.5.4.2.1, 5.5.4.2.2, 5.5.4.2.3; 5.1.3	For **ASHRAE-compliance-path** jobs, provide **fenestration area calculation path** to prove (a) Vertical fenestration area ≤ 40%, Skylight area ≤ 3%, or (a) Vertical fenestration area > 40%, and/or Skylight area > 3% with compliance through Section 5.6-Building envelope trade-off or Energy modeling, and (c) For spaces defined in 5.5.4.2.3, the Minimum skylight area provided
EN-v.4 Opaque Envelope Components	C402, 5.5.3, 1 RCNY §5000-01 (g)(1); C502.1, C503.1, C503.2, C503.3, C503.3.1, C505.1, 5.1.3	Provide **assembly details for opaque envelope components** to demonstrate assembly R-values and/or U-factors meet requirements: (b) Walls - above-grade and below-grade (c) Floors and Slab-on-Grade (d) Opaque doors (e) Spandrel panels (f) Parapets, Balconies, Slab edges, etc. for all new construction, additions, and alterations as applicable
EN-v.4.1 Envelope Components for Alterations	101.2.5, C503.1, C504.2, 5.1.3	For Alteration areas where **Alteration-specific exceptions/allowances** are applied, provide detailed drawing/description for each Alteration area to support the corresponding row in Tabular Analysis.
EN-v.5 Fenestration Assembly	C402.4, C402.5.2, 5.5.4.3, 5.5.4.4, 5.4.3, 1 RCNY §5000-01 (g)(1); C503.3.2, C503.3.3, C503.3.4, 5.1.3	Provide **fenestration (window/door/skylight) schedules** specifying each assembly type's (a) Assembly U-factor (b) SHGC/ VT, and (c) Air leakage rate to meet the requirements. For buildings with height > 95' above grade, indicate **demarcation lines at 95**' above grade in building elevations.

Objection	Code Sections	Objection Details
EN-v.6 Thermal Bridges	C402.6, 5.4.4, 1 RCNY §5000-01 (g)(1)(iii);	Document individual **thermal bridges** with supporting details: (a) Clear field thermal bridges -- relating to wall, roof, floor assembly details (b) Point thermal bridges -- for each point bridge > 12 sq.in. (c) Linear thermal bridges -- in tabular format
EN-v.7 Air Leakage Limits	C402.5, 5.4.3; C503.3, 5.1.3	Specify mandatory provisions to **limit air leakage** in the building thermal envelope: (a) General envelope design/construction to ensure a continuous air barrier (b) Dampers in air intake/exhaust/shaft vents (c) Vestibules and Loading dock weatherseals
EN-v.8 Air Leakage Testing	C402.5.1.3, C408.4, 5.4.3.1.3, 5.7.5, 1 RCNY §5000-01 (g)(5)(iv); C503.3, C401.2.1, 5.1.3	For new buildings or additions ≥ 10,000 sf and < 50,000 sf, and height ≤ 75', clearly state on drawings: (a) **Air leakage testing** is required, (b) Air leakage testing shall be performed in accordance with ASTM E779 at 75 Pa, (c) Testing results shall not exceed 0.4 cfm/sf, and (d) Provide Field inspection checklist for Air barrier commissioning (C408.4.1/ 5.7.5.1)
EN-v.9 Air Barrier Continuity Plan Testing/Inspection	C402.5.1.3, C408.4, 5.4.3.1.3, 5.7.5, 1 RCNY §5000-01 (g)(5)(iv); C503.3, 5.1.3	For new buildings or additions ≥ 10,000 sf and < 50,000 sf, and height > 75', or for new buildings or additions ≥ 50,000 sf, clearly state on drawings: (a) Testing/inspection per Air Barrier Continuity plan is required, (b) **Air Barrier Continuity Plan** on drawings to specify: (b.1) Air sealing details keyed to building thermal boundary plan/section diagrams (b.2) Sealing methods/materials (b.3) Testing methods/standards (c) Provide Field inspection checklist for Air barrier commissioning (C408.4.1/ 5.7.5.1)
EN-v.10 Mechanical Equipment Penetrations	C402.1.4.3, 5.5.3; C503.3, 5.1.3	Account for the wall area of **mechanical equipment penetrations** as a separate wall assembly of U-0.5 when the total penetration areas exceed 1% of the total above-grade wall area.
EN-v.11 (a) Fenestration Orientation	5.5.4.5	For ASHRAE-compliance-path jobs, provide documentation to prove that West-oriented and East-oriented **fenestration areas and solar heat gains** are designed not to exceed the code-prescribed limits.

Objection	Code Sections	Objection Details
EN-l [Lighting & Power]		
EN-l.1 Lighting Power Density	C405.3, C405.4, C405.5, C405.1, 9.2.2.3, 9.4.2, 9.4.3, 9.1.1, 9.4.4, 1 RCNY §5000-01 (g)(3)(i)(A); C502.2.6.1, C502.2.6.2, C503.6, 9.1.2	Provide complete light fixture layout plans, light fixture schedules (including fixture wattage and counts) and notes not to exceed the maximum **lighting power density allowance** for: (a) Interior lighting power (b) Exterior lighting power (c) Exit signs, and (d) High-efficacy Lamps/Luminaires for dwelling units: Minimum 90% of Lamps with min. 65 lumens/watt, Or Minimum 90% of Luminaires with min. 45 lumens/watt
EN-l.1.1 Lighting & Other Equipment for Alterations	101.2.5, C503.1, C503.6, C504.2, 8.1.4, 9.1.2, 10.1.1.3	For Alteration areas where **Alteration-specific exceptions/allowances** are applied, provide detailed drawing/description for each Alteration area to support the corresponding row in Tabular Analysis.
EN-l.2 Lighting Controls	C405.2, C405.4, C405.5, C405.1, C408.3, 9.4.1.1, 9.4.1.2, 9.4.1.3, 9.4.1.4, 9.4.3, 9.7.3.1, 1 RCNY §5000-01 (g)(3)(i)(A); C502.2.6, C503.6, 9.1.2	Specify **interior/exterior lighting controls** as required, and key them to lighting plans to account for: (a) Local controls (b) Occupancy sensor controls (with Manual-On unless permitted otherwise) (c) Bi-level controls (d) Daylight-responsive controls (e) Full-off/partial off (f) Scheduled shutoff for interior lighting (g) Photocell and time switches for Exterior lighting Provide **Lighting/Power Controls Narrative** to - list ALL proposed interior and outdoor spaces of the building, and - specify mandatory requirements in lighting/ power controls for each space. Direct on drawings that '**Functional Testing of Lighting Controls**' is required.
EN-l.3 Electrical Energy Metering	C405.6, 8.4.5; 8.1.3, 8.1.4	Specify separate **electrical meter** or sub-meter for each dwelling unit and/or tenant space for buildings > 25,000 sf with tenant spaces > 5,000 sf.
EN-l.4 Elevator Cab	C405.8.1, 10.4.3; C503.6, 10.1.1.2, 10.1.1.3	Specify **elevator cab** lamp efficacy, ventilation power limit, and controls on fans & lighting systems to meet requirements.
EN-l.5 (a) Electrical Energy Monitoring	8.4.3	Specify **electrical energy monitoring** devices in new buildings to separately measure/record total electrical energy, HVAC systems, interior lighting, exterior lighting, and receptacle circuits.
EN-l.6 Whole-Building Energy & Electrical Monitoring	C405.11, C405.12, 10.4.5	Specify **whole-building energy & electrical monitoring** devices in new buildings to individually monitor energy use of each energy type supplied by utility, energy provider, or plant outside the building.
EN-l.7 Hotel/Motel Sleeping Units	C405.2.4(2), 9.4.1.3(b); C503.6	For **hotel/motel sleeping units** and guest suites, specify master control device that automatically switches off all installed luminaires and switched receptacles within 20 minutes (ECC)/ 15 minutes (ASHRAE) of occupants' leaving the room.

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EN-I.8 (a) Parking Garage Lighting Controls	9.4.1.2	Specify **parking garage lighting controls** as required.
EN-I.9 (a) Automatic Receptacle Controls	8.4.2; 8.1.3, 8.1.4	Specify **automatic receptacle controls** on minimum 50% of receptacles in offices, conference rooms, copy/print rooms, break rooms, individual workstations, and classrooms.
EN-I.10 Elevators, Escalators, & Moving Walks	C405.8.1.1, C405.8.2, 10.4.3.5, 10.4.4, 1 RCNY §5000-01 (g)(3); 10.1.1.3	Specify **elevators, escalators, and moving walks** with required efficiency, controls, and regenerative drives.

Objection	Code Sections	Objection Details
EN-m [Mechanical & Plumbing]		
EN-m.1 Commissioning	1 RCNY §5000-01 (g)(5)(iii), C408, 6.7.2.3; C502.2.3.1, C502.2.4.1, C503.4.2, C503.5.1, Exceptions to 6.7.2.3	Specify **system commissioning requirements** on drawings: (a) A statement indicate whether commissioning is required, and (b) Total installed heating (HVAC and SWH combined) capacity and total installed cooling capacity specified by the commissioning statement (plus, renewable energy capacity if proposed)
EN-m.2 Equipment Sizing Statement	C403.1.1, C403.3.1, 6.4.2; C502.2.3, C503.4	Provide equipment sizing statement that references procedures in the **ANSI/ASHRAE/ACCA Standard 183** for determining heating and cooling design loads.
EN-m.3 Efficiency Values/ Performance Ratings	C403.3.2, C404.2, C405.10, 6.4.1, 6.8.1, 7.4.2, 10.4.6; C502.2.3, C502.2.4, C503.4, C503.5, 6.1.1.2, 6.1.1.3, 7.1.1.2, 7.1.1.3	Specify **efficiency values/ performance ratings** in equipment schedules and/or notes. Values must be noted in the measurement unit prescribed on the corresponding equipment code table: (a) HVAC systems (b) Service water heating systems (c) Refrigeration equipment (d) Commercial kitchen equipment
EN-m.3.1 (a) HVAC & Service Water Heating for Alterations	6.1.1.3, 6.4.3.10.1, 7.1.1.3	For Alteration areas where **Alteration-specific exceptions/allowances** are applied, provide detailed drawing/description for each Alteration area to support the corresponding row in Tabular Analysis.
EN-m.4 System Controls	C403.4, C404.3, C404.6, 6.4.3, 6.5.2, 7.4.6, 7.4.4.2, 7.4.4.3, 7.4.4.4; C502.2.3, C502.2.4, C503.4, C503.5, 6.1.1.2, 6.1.1.3, 7.1.1.2, 7.1.1.3	Specify **system controls** on drawings as required including: (a) Thermostatic zone control (b) Deadband and Setpoint control (c) Outdoor temperature setback control (d) Off-hour controls: automatic setback/ shutdown controls, optimum start controls (e) Heat pump supplementary heat (f) Service water heating equipment (non-circulating/ circulating) temperature maintenance and circulation controls, e.g., heat traps, circulation pump controls, heat trace systems, etc. (g) System **controls narrative** provided to describe function and operation of each mandatory control system
EN-m.5 Shutoff Dampers	C403.7.7, 6.4.3.4; C502.2.3, C503.4, 6.1.1.3.1	Specify **motorized shutoff dampers** for stairway enclosures, elevator shaft vents, and other outdoor air intakes and exhaust openings. (a) Indicate individual damper locations on plans and/or riser diagrams. (b) Specify air leakage rate as required.
EN-m.6 Duct, Plenum and Piping Insulation and Sealing	C403.11, C404.4, 6.4.4.1, 6.4.4.2, 6.8.3, 7.4.3; C502.2.3, C502.2.4, C503.4, C503.5, 6.1.1.3.4, 6.1.1.3.5	Specify **duct, plenum and piping insulation and sealing** requirements for HVAC and service water heating systems on drawings.

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EN-m.7 Economizers	C403.5, 6.4.3.12, 6.5.1; C502.2.3, C503.4.1, 6.1.1.3.3	Specify **economizers** in equipment schedules as required: (a) On individual fan-cooling units ≥ 270 kBtu/h for Group R occupancies, and units ≥ 54 kBtu/h for all other occupancies (b) For ECC-compliance-path jobs only: Total capacity of Economizer-exempt fan-cooling units not exceeding - 20% of the total fan-cooling units capacity in the building, or - 1,500 kBtu/h for Group R occupancies, or - 300 kBtu/h for all other occupancies, whichever is greater (c) With economizer FDD system, high limit shut-off control, and integrated cooling controls (d) With exception of cooling units demonstrating - 20% efficiency improvement on ECC-compliance-path jobs, or - 42% efficiency improvement on ASHRAE-compliance-path jobs
EN-m.8 Hot Water Boiler Controls	C403.3.4, C403.4.1.5, C403.4.3, 6.5.4.1, 6.5.4.4; C502.2.3, C503.4, 6.1.1.3.1	Specify **hot water boiler controls** on drawings as required including: (a) Modulating burner and/or Minimum turndown ratio (b) Outdoor temperature setback controls
EN-m.9 Energy Recovery Ventilation	C403.7.4, 6.5.6.1; C502.2.3, C503.4	Specify **energy recovery ventilation (ERV)** system as required, and identify exhaust air energy recovery effectiveness ratio values in equipment schedules.
EN-m.10 Demand Control Ventilation	C403.7.1, 6.4.3.8; C502.2.3, C503.4, 6.1.1.3.1	Specify **demand control ventilation (DCV)** in the spaces/systems as required.
EN-m.11 Hotel/Motel Sleeping Units – HVAC Control	C403.7.6, 6.4.3.3.5; C502.2.3, C503.4, 6.1.1.3.1	Specify **HVAC controls for hotel/motel sleeping units** by automatic setback for temperature and ventilation control during unoccupied hours, or by captive key card system.
EN-m.12 Enclosed Parking Garage Ventilation Controls	C403.7.2, 6.4.3.4.5; C502.2.3, C503.4, 6.1.1.3.1	Specify **ventilation controls for enclosed parking garage** as required.
EN-m.13 Pools and Spas - Energy Conserving Measures	C404.9, C404.10, C404.11, 7.4.5; C502.2.4, C502.2.5, C503.5, 7.1.1.3	Specify **energy conserving measures for pools and spas** as required, e.g., heater controls, time switches, pool covers, etc.
EN-m.14 Fan Power Limits	C403.8, 6.5.3.1; C502.2.3, C503.4, 6.1.1.3.1	Specify **fan power limits** for systems with total fan power > 5hp: (a) To meet the allowable fan system bhp or allowable nameplate motor hp, or (b) To have fan efficiency grade of ≥ 67
EN-m.15 (a) DDC Controls and Display	6.4.3.10; 6.4.3.10.1	Specify **DDC controls and display** systems for new buildings with chilled-water or hot-water plants of ≥ 300 kBtu/h, or fan systems bhp of ≥ 10hp.
EN-m.16 (a) Chilled-Water Plant Monitoring	6.4.3.11; 6.4.3.11.1	Specify **chilled-water plant monitoring** devices for (a) water-cooled chilled-water plants of > 1,000 tons peak cooling capacity, or (b) air-cooled chilled-water plants of > 570 tons peak cooling capacity

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EN-m.17 Hydronic System Controls	C403.4.3, 6.5.2.2, 6.5.4.2, 6.5.4.3; C502.2.3, C503.4, 6.1.1.3.1	Specify **hydronic system controls** on drawings as required.
EN-m.18 High Efficiency Gas Boiler Systems	C403.3.5, 6.4.1.6; C502.2.3, C503.4, 6.1.1.3.1	For high-efficiency gas boiler systems, specify **return water temperature of ≤ 120 °F.**
EN-m.19 Complex System Controls	C403.6, C403.9.5, C403.3.3, 6.5.3.2, 6.5.3.3, 6.5.3.4, 6.5.6.2, 6.5.9; C502.2.3, C503.4, 6.1.1.3.1	Specify **complex system controls** on drawings, e.g., VAV systems serving multiple zones, condenser heat recovery installation for service water heating, hot gas bypass limits, etc.

Objection	Code Sections	Objection Details
EN-r [Residential]		
EN-r.1 Opaque Envelope Components	R402, 1 RCNY §5000-01 (g)(1); R502.1.1.1, R503.1.1	Provide **assembly details** for opaque envelope components to demonstrate assembly R-values and/or U-factors meet requirements: (a) Ceiling (Roof) (b) Wall - Framed and Mass wall (c) Floor (d) Basement Wall (e) Slab (f) Crawl Space Wall (g) Others - e.g., Parapet, Balcony, Slab Edge for all new construction, additions, and alterations as applicable
EN-r.1.1 Alteration Components	101.2.5, R403.3.1, R504.2, R503.1.4	For Alteration areas where **Alteration-specific exceptions/allowances** are applied, provide detailed drawing/description for each Alteration area to support the corresponding row in Tabular Analysis.
EN-r.2 Fenestration Assembly	R402.1.2, R402.1.4, R402.3, R402.4.3, 1 RCNY §5000-01 (g)(1); R502.1.1.1, R503.1.1, R503.1.1.1	Provide **fenestration (window/door/skylight) schedules** specifying each assembly type's (a) Assembly U-factor (b) SHGC, and (c) Air leakage rate to meet the requirements.
EN-r.3 Air Leakage Limits	R402.4; R502.1.1.1	Provide mandatory notes/details to **limit air leakage** in the building thermal envelope including: (a) Air-barrier and sealing criteria for general envelope components (b) Fuel-burning appliances isolated or direct-vented (c) Min. R-10 insulation and air sealing at tenant separation walls (d) Recessed luminaires meeting air leakage rate limit and sealing requirements (d) Tight-fitting noncombustible fireplace doors
EN-r.4 Air Leakage Testing	R402.4.1; R502.1.1.1, R402.4.1.2, R402.4.1.3	Specify on drawings: (a) **Air leakage testing** shall be performed in accordance with RESNET/ICC 380, ASTM E779 or ASTM E1827, (b) Testing results shall not exceed 3 air changes per hour(ACH) at 50 Pa, and (c) Applicable testing and reporting guidelines as prescribed in the code section.
EN-r.5 Elements in Thermal Envelope	R402.2; R502.1.1.1, R503.1.1	Specify requirements for **elements in thermal envelope** including: (a) Access hatches and doors weather-stripping and insulation (b) Thermally isolated sunroom insulation and fenestration
EN-r.6 Thermal Bridges	R402.6, 1 RCNY §5000-01 (g)(1)(iii)	Document individual **thermal bridges** with supporting details: (a) Clear field thermal bridges -- for those not identified in Appendix A of ASHRAE 90.1 (b) Point thermal bridges -- for each point bridge > 8 sq.in. (c) Linear thermal bridges -- in tabular format

Objection	Code Sections	Objection Details
EN-r.7 Envelope Insulation Materials/Methods	R303	Specify **envelope insulation materials/methods** to meet requirements including: (a) Blown or sprayed roof/ceiling insulation in the attic with clear marker installation (b) Protection of exposed foundation insulation
EN-r.8 Permanent Certificate Requirements	1 RCNY §5000-01 (g)(4), R401.3; R501.4	Specify on drawings **permanent certificate requirements** including where it shall be posted and what information to be listed.
EN-r.9 Electric Vehicle Service Equipment Capable	R404.3; R501.4	Provide **electric vehicle service equipment (EVSE)** capable measures on drawings by specifying: (a) 208/240V 40-amp outlet per dwelling unit, or (b) panel capacity and conduit for future installation of 208/240V 40-amp outlets
EN-r.10 Equipment Sizing Statement	R403.7, 1 RCNY §5000-01 (g)(5); R502.1.1.2, R503.1.2	Provide equipment sizing statement referencing compliance with ACCA Manual S based on **ACCA Manual J**.
EN-r.11 Equipment System Efficiency & Controls	R403.7, R403.8, 1 RCNY §5000-01 (g)(2); R502.1.1.2, R503.1.2	Provide complete heating and cooling equipment schedules to specify **equipment system efficiency** (and controls as applicable): (a) Efficiency rating in compliance with federal minimum requirements, and/or (b) Systems serving 3 or more dwelling units in compliance with Sections C403 and C404 in lieu of R403
EN-r.12 Programmable Thermostat	R403.1.1; R502.1.1.2, R503.1.2	Provide at least one **programmable thermostat** for each separate heating and cooling system.
EN-r.13 Boilers with Setback Control	R403.2; R502.1.1.2, R503.1.2	Specify **hot water boilers** with outdoor temperature setback control.
EN-r.14 Piping Insulation & Sealing	R403.4, R403.5.3; R502.1.1.2, R502.1.1.3, R503.1.2, R503.1.3	Specify mandatory **piping insulation and sealing** requirements: (a) Required minimum thickness insulation for mechanical system piping (b) Min. R-3 insulation for service hot water system piping
EN-r.15 Duct System Location	R403.3, R403.3.7; R502.1.1.2, R503.1.2	Ensure **ducts** in new buildings and additions are located **in conditioned space**.
EN-r.16 Service Hot Water Supply Piping	R403.5.5	Specify **service hot water supply piping** to meet one of the requirements: (a) Maximum allowable pipe length method (b) Maximum allowable pipe volume method (c) Drain water heat recovery units (d) Recirculation systems
EN-r.17 Ventilation Design	R403.6.2	Specify for dwelling units in new buildings. (a) Supply and exhaust ventilation with **HRV or ERV**, or (b) **Whole-house Balanced Ventilation** system with interlocked operation of supply and exhaust fans.

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EN-r.18 Ventilation Control & Efficiency	R403.6; R502.1.1.2, R503.1.2	Specify **ventilation control and efficiency** to meet requirements including: (a) Automatic or gravity dampers for outdoor air intakes and exhausts (b) Ventilation fans met minimum efficacy requirements
EN-r.19 Lighting and Electrical Power	R404; R502.1.1.4, R503.1.4	Provide lighting and power plans, schedules, and/or notes to meet the requirements including: (a) **High-efficacy lamps or luminaires**: Minimum 90% of fixtures using - lamps with min. efficacy of 65 lumens/watt, or - luminaires with min. efficacy of 45 lumens/watt (b) Separate electrical meter or sub-meter for each dwelling unit

NOTE: In this reference document, selected Energy Code provisions have been generalized, summarized, rephrased, and/or highlighted. This reference document is intended: 1) To provide general guidance for the job applications seeking compliance with the 2020 NYCECC; 2) Not to replace or represent the entire 2020 NYCECC and related regulations of the City of New York and the NYC Department of Buildings; and 3) Not to provide complete compliance solutions for any particular type of job or work. Comprehensive mandates, applicability, exemptions, exceptions and options are found in the 2020 NYCECC and related regulations of the City of New York and the NYC Department of Buildings.

NYC Energy Conservation Code https://www.nyc.gov/site/buildings/codes/energy-conservation-code.page
2020 NYCECC https://www.nyc.gov/site/buildings/codes/2020-energy-conservation-code.page